Assessment Assumptions, Baseline, Sensitivity Analysis & Budgeting

Practical guidance for preparing and testing a credible 12-month SME projection.

This guide explains how to prepare the assessment baseline, use Sensitivity Analysis to test uncertainty, and turn the assessment projection into a managed 12-month budget. It supports government-programme assessment and management discussion; it is not a substitute for audited accounts, professional tax advice or a guarantee of future performance.

> **Recommended sequence:** capture verified opening facts, enter reasonable baseline assumptions, review readiness, confirm the baseline, test uncertainty with Sensitivity Analysis, and then create the assessment budget for management approval and monitoring.

1. Opening Position

Setting	How it supports projections
Opening stock, debtors and creditors	Read-only totals calculated from active Sales Categories. Update the underlying category records rather than entering duplicate totals.
Opening bank balance	Sets the initial cash available. Every projected receipt and payment builds from this amount.
Overdraft interest	Provides the annual financing rate applied where projected cash becomes overdrawn.
Gross assets, accumulated and projected depreciation	Read-only totals calculated from active asset records and the asset depreciation engine.
Other current assets and liabilities	Captures prepayments, non-trade receivables, accruals and short-term obligations not represented by trade debtors or creditors. These remain on the projected balance sheet unless reclassified outside this lightweight model.
Opening VAT payable	VAT already owed at the start date and treated as a Month 1 cash outflow.
Opening VAT refundable	A confirmed VAT refund receivable at the start date and treated as a Month 1 cash inflow.

Evidence: recent bank statement, stock count, management accounts, aged debtors/creditors, asset register and loan statements. A genuine zero should be confirmed rather than left unknown.

2. Sales Categories

Setting	Projection use

Category and income type	Separates true Sales from sundry income, grants, disposals and other income. Only true Sales should drive gross profit, stock, debtors and break-even measures.
Markup	Supports the category cost-of-Sales estimate and gross-margin calculation when purchase/stock information is incomplete.
Opening stock	Assigns starting stock to the relevant Sales Category and feeds opening working capital.

3. Debtors & Creditors Profiles

Percentage mode Enter the expected distribution across ageing buckets. The 0–30 day bucket balances the total to 100%.

Actual values mode Enter known balances such as R90,000 in 0–30 days and R35,000 in 31–60 days. The application converts each bucket to a percentage of the total and saves those percentages for projections.

Setting	Projection use
Debtor profile	Determines when credit Sales are expected to be collected, shifting cash receipts into later months.
Creditor profile	Determines when supplier purchases are paid, shifting cash payments into later months.
Bad debt	Reduces the collectible portion of credit Sales and highlights collection risk.
Average payment days	Provides a practical reasonableness indicator for the captured collection profile.
Include VAT	Clarifies whether the entered ageing balances include VAT so cash and VAT interpretation remain consistent.

4. Sales & Stock Assumptions

Setting	Projection use
Sales growth and start month	Defines the baseline percentage change and the first month affected. “Month 4” means Months 1–3 remain at the entered baseline.
Cash Sales percentage	Identifies Sales collected in the month of sale; the balance follows the debtor profile.
Bad debts	Provides a baseline loss allowance for uncollectible credit Sales. Reconcile it with the profile setting.
Fallback cost of Sales	Used only when category markup or detailed stock/purchase data cannot provide a reliable cost basis.
Minimum closing stock	Defines a simple stock buffer and supports purchase requirements and liquidity review.

5. Expenses, VAT & Director Wages

Setting	Projection use
VAT rate	Applies to taxable Sales, purchases and expenses. Confirm registration status and the legally applicable rate.

VAT payment frequency	Controls whether net VAT cash settlement is expected monthly or every two months.
Expense increase and start month	Escalates operating expenses from the selected month without changing earlier baseline months.
Contingency	Adds a modest allowance for foreseeable but unallocated operating costs. It should not conceal known expenses.
Director wages	Number of directors multiplied by monthly wage becomes the recurring management remuneration cash outflow.
Director wage increase	Applies the expected change to director remuneration during the projection period.

6. Funding, Cash Reserve & Assets

Setting	Projection use
Loan balance, rate, term and instalment	Drive financing balances, interest expense and monthly repayment cash outflows. The loan start month determines when a new facility begins affecting the projection.
Minimum cash balance	Represents management's preferred liquidity reserve. Falling below it should trigger a warning, not be treated as additional spending capacity.
Asset opening cost and accumulated depreciation	Establish net book value at the start of the projection.
Acquisition month	Determines when a planned asset enters the balance sheet and begins affecting cash and depreciation.
Depreciation rate and method	Determine the periodic non-cash depreciation charge and closing asset value.
Disposals and movements	Adjust asset values, disposal proceeds and gains/losses in the relevant month.

7. Readiness, Notes & Confirmation

The readiness score checks whether essential opening facts, Sales Categories, monthly Sales, working-capital profiles, expenses, VAT, loans, asset depreciation and baseline assumptions are available. **Incomplete** means a required check failed. **Ready with warnings** means core data is available but an advisory item remains. **Ready** means the minimum automated checks pass.

Assumption notes should record the source, date, owner and reason for each material estimate—for example, "based on signed orders" or "average of the last six VAT returns."

Confirmation records that the baseline has been reviewed and is reasonable for the assessment. It does not prove that the forecast will occur.

> **Important cautions:** avoid double-counting assumptions already represented by monthly source data; use fallback cost of Sales only when required; distinguish Sales from grants or disposals; keep VAT treatment consistent; and rerun readiness after changing material source data.

8. Using Sensitivity Analysis

Sensitivity Analysis tests how the approved 12-month projection responds when important assumptions change. It does not edit Sales, expenses, assets, loans or any other source record. Every analysis begins with a fresh copy of the authoritative baseline and applies temporary adjustments in memory.

> **Purpose:** use the module to understand downside risk, test opportunities, estimate funding pressure and explain which assumptions have the greatest influence on viability. It should support professional judgement rather than replace it.

8.1 Before starting

1. Open the correct assessment and confirm that its opening position and monthly source data are current.
2. Review Projection Readiness and resolve material missing information.
3. Confirm that the baseline projection is reasonable before testing alternatives.
4. Select **Sensitivity Analysis** immediately after **Assessment Reports** in the assessment menu.

If Sales or expense information is incomplete, the page displays a **Draft baseline** warning. Analysis can still be run, but its conclusions must be treated as provisional.

8.2 Scenario Builder

The Scenario Builder combines one or more changes into a practical alternative case. Start with a preset or enter a name, description and individual adjustments.

Control	How to use it
Scenario name and description	Use a clear name and explain the event being tested, such as “Municipal contract delayed” or “Supplier increase from Month 4.”
Driver	Select the financial assumption to change.
Adjustment	Enter a positive value for an increase or delay and a negative value for a decrease or acceleration, according to the driver.
Effective month	Select Month 1–12 when the change begins. Earlier months remain unchanged.
Duration	Select **Ongoing** for the rest of the projection or clear it and enter a limited number of months.
Add adjustment	Add the change to the scenario. Several adjustments may be combined before calculation.
Run scenario	Calculates the scenario and compares its main results with the baseline.

Scenario presets

Preset	Question it helps answer
Sales Decline	What happens if demand is weaker than expected?
Cost Inflation	Can the enterprise absorb increases in inputs and overheads?

Slow Collections	How does a customer-payment delay affect cash?
Interest Rate Increase	How sensitive is profitability to higher finance costs?
Asset Acquisition	Can the planned capital expenditure be funded?
Combined Downside	What happens when lower Sales, higher costs and slower collections occur together?
Growth Opportunity	Does additional Sales justify the extra operating expenditure?

Presets are starting points only. Review and amend every adjustment so that it reflects the enterprise being assessed.

8.3 Understanding sensitivity drivers

Driver	Meaning and interpretation
Sales / revenue	Changes projected true Sales by the selected percentage. A value of −10% tests Sales that are 10% below baseline.
Cost of Sales — relative	Changes baseline cost of Sales by a percentage. A value of +10% makes the existing cost amount 10% higher.
Cost of Sales — percentage points	Changes cost of Sales as a share of revenue. If baseline cost is 60% of Sales, +5 percentage points tests 65%.
Operating expenses	Changes recurring operating expenses by the entered percentage for the selected period.
Bad debt	Adds percentage points to the uncollectible portion of Sales and reduces expected customer receipts.
Debtor collection timing	Moves projected customer receipts by whole months. A positive value delays receipts; a negative value accelerates them.
Creditor payment timing	Moves projected supplier payments by whole months. A positive value delays payment; consider whether suppliers would permit it.
Loan interest	Changes projected loan-interest expense by a relative percentage. It does not amend contractual principal repayments.
Asset purchases	Changes baseline projected asset-purchase cash outflows by the entered percentage.
Minimum cash reserve	Sets the management buffer used to identify funding shortfalls. It is a risk threshold, not an additional expense.

> **Percentage versus percentage points:** these are different. Increasing a 60% cost-of-Sales ratio by 10% produces 66%; increasing it by 10 percentage points produces 70%.

8.4 Reading Scenario Builder results

Measure	What to assess
Revenue	Total true Sales over Months 1–12 after scenario adjustments.
Gross profit	Revenue less cost of Sales. Review both the amount and gross-margin percentage.
Profit before tax (PBT)	Profit after operating expenses, depreciation and interest, before tax.
Closing cash	Cash remaining at Month 12. A positive closing balance can still conceal an earlier shortfall.
Minimum cash	The lowest monthly closing cash balance and the month in which it occurs.

Variance from baseline	The amount by which the scenario result improves or deteriorates relative to the approved baseline.

The deterministic explanation lists the largest adjustments and the resulting movement in profit and closing cash. Expand **Calculation scope** to review modelling limitations.

8.5 Sensitivity Results

This tab changes one driver at a time while holding all others at baseline. It is the best place to identify which assumption has the greatest effect.

1. Select the driver.
2. Enter the minimum and maximum adjustment.
3. Enter the step between test points. For example, −30 to +20 with a step of 5 tests −30%, −25%, −20% and so on.
4. Select the effective month and choose **Run range**.
5. Compare the baseline row with each test point and review the status colour and warning flags.

The chart shows closing cash across the range. The table adds revenue, gross margin, EBITDA, PBT, minimum cash, funding gap, current ratio and interest cover. Use **CSV** to download the detailed results for further review.

8.6 Scenario Comparison

This tab compares the baseline with up to three named scenarios. Select the scenarios to display their KPI variances and M1–M12 revenue, PBT and closing-cash movements.

- Review amount and percentage variances, not only final totals.
- Identify the first month in which cash or profit becomes unacceptable.
- Compare the cause descriptions to distinguish operational risk from financing risk.
- Export the comparison to CSV when the evidence must be shared or retained.

8.7 Risk & Break-even

Enter management's minimum cash reserve and recalculate. The module progressively tests the baseline to estimate practical limits.

Threshold	Interpretation
Maximum Sales decline	The first tested decline that produces a loss or breaches the selected cash reserve.
Maximum operating-expense increase	The first tested increase that produces a loss or breaches the reserve.
Minimum gross margin	An approximate annual margin required to cover operating expenses, depreciation and interest.
Collection-delay tolerance	The first whole-month customer-payment delay that breaches the reserve.
Minimum projected cash	The lowest monthly cash balance in the baseline and the month in which it occurs.

Findings are prioritised and paired with a practical management action. They are prompts for investigation and must be considered together with contracts, security, funding terms and management capacity.

8.8 Status colours

Status	Meaning	Suggested response
Green	No tested key threshold is breached.	Continue monitoring and retain contingency plans.
Amber	A cash reserve or coverage warning exists.	Investigate the affected month and agree a mitigation before approval.
Red	The scenario produces negative cash or a projected loss.	Do not rely on the case without corrective action, additional funding or revised assumptions.

8.9 Printing and exports

Select **Print / Save PDF** to open the browser print dialogue and save the active analysis view as a PDF. The Sensitivity Results and Scenario Comparison tabs also provide CSV exports. Record the assessment, baseline date, scenario name, adjustments and calculation date when sharing results.

8.10 Important calculation limitations

> - Scenarios are held in memory for the current page session and are not saved to the database.

- The baseline remains unchanged; update source data on the appropriate assessment page when a change is intended to become permanent.
- Existing VAT schedules remain at baseline during sensitivity overlays.
- Collection or supplier payments shifted beyond Month 12 fall outside the analysis horizon.
- Asset-purchase sensitivity changes the projected purchase cash outflow; it does not rebuild a new detailed asset register or depreciation schedule.
- Thresholds are tested estimates and may be rounded to the selected increment or whole month.
- Sensitivity results show what could happen under stated assumptions; they do not assign probabilities or predict which scenario will occur.

8.11 Recommended sensitivity review

1. Run the baseline and confirm that no material data is missing.
2. Run a Sales range to understand demand risk.
3. Run cost-of-Sales and operating-expense ranges to understand margin pressure.
4. Test slower collections and review the minimum-cash month.
5. Run the Combined Downside preset and adapt it to the enterprise.
6. Compare baseline, optimistic and pessimistic cases.
7. Set the required cash reserve and review risk thresholds.
8. Document management actions, supporting evidence and unresolved uncertainties.
9. Export or print the final analysis for the assessment record.

9. Using the Assessment Budget

The Budgeting module converts the current assessment projection into a controlled 12-month management budget. Every budget belongs to one assessment and uses the same Month 1–12 period and assessment start date.

> **Important distinction:** the assessment projection is the source and comparison baseline. The budget is a separate planning record. Editing the budget does not change Sales, expenses, stock, assets, loans, VAT, cash-flow calculations or any other assessment source record.

9.1 Before creating a budget

1. Open the correct assessment and verify the case number displayed in the assessment layout.
2. Complete the material Sales, stock, expense, asset, loan, VAT and working-capital information.
3. Review Projection Readiness and resolve material missing data.
4. Generate the assessment reports and confirm that projected profit and cash flow are reasonable.
5. Use Sensitivity Analysis where appropriate to understand the main risks before committing to targets.

A budget may still be created from incomplete assessment information, but the imported amounts will reflect those omissions. Correct the assessment first whenever the missing information is material.

9.2 Opening the module

1. Open the required assessment.
2. Select the shortcuts menu in **MainLayout_Assessment**.
3. Select **Budgets — Assessment Budget & Variance**.
4. The application opens the budget belonging to that assessment. A budget from another assessment cannot be displayed through this route.

9.3 Creating the first budget

Each assessment supports one active budget. If no budget exists, enter a meaningful budget name, add optional planning notes, and select **Create from assessment projection**.

ViabilityIQ copies the assessment's current 12-month projection into two sets of values:

Value	Purpose
Projection	A dated snapshot of the assessment result used as the comparison baseline.
Budget	The management target. It initially equals the projection and can then be edited independently.

The snapshot includes revenue, other income, cost of goods sold, operating expenses, depreciation, interest, receipts, supplier payments, VAT payments, loan payments, asset purchases and calculated profit and cash-flow totals.

9.4 Editing the monthly planning matrix

Editable rows contain white monthly input fields. Enter the planned amount for each Month 1–12. Amounts are displayed to two decimal places.

Section	Typical editable lines	Calculated lines

Profit & Loss	Revenue, other income, cost of goods sold, operating expenses, depreciation and interest expense.	Gross profit, EBITDA, EBIT and profit before tax.
Cash Flow	Sales receipts, other receipts, supplier payments, cash operating expenses, VAT payments, loan payments and asset purchases.	Total inflows, total outflows, net cash movement and closing cash balance.

Calculated rows cannot be entered manually. They update automatically when an editable amount changes and are recalculated again by the server when the draft is saved.

> **Profit and cash are different:** changing budgeted revenue does not automatically change Sales receipts. Likewise, depreciation affects profit but is not a cash payment. Review both sections and keep related assumptions consistent.

9.5 Reading the three display modes

Mode	Meaning	Recommended use
Budget	Shows the current management targets. Editable lines can be changed while the budget is Draft.	Prepare and revise the budget.
Projection	Shows the most recently imported assessment projection snapshot.	Confirm the financial model on which the budget was based.
Variance	Shows Budget minus Projection for each month and total.	Identify deliberate departures from the assessment result and document the reason.

A positive variance means the budget amount is higher than the projection; a negative variance means it is lower. Whether that is favourable depends on the line. Higher revenue may be favourable, while higher costs or payments may be unfavourable.

9.6 KPI cards and alerts

Indicator	Interpretation
Budget revenue	Total budgeted true revenue for Months 1–12 and its difference from projection.
Profit before tax	Budgeted profit after operating costs, depreciation and interest, before tax.
Net cash movement	Total budgeted cash inflows less cash outflows over the 12 months.
Closing cash	Month 12 cash position after applying monthly net cash movements to the assessment opening bank balance.

Budget alerts highlight a closing funding shortfall, loss-making months and a material deterioration in profit relative to the assessment projection. Alerts are management prompts; they do not prevent saving or submitting the budget.

9.7 Saving and refreshing the projection

Select **Save draft** after changing budget values, the name or planning notes. The page marks unsaved changes and will not submit them until they are saved.

Select **Refresh projection** when assessment source records have changed and the comparison baseline must be updated. Refreshing:

- recalculates the live assessment projection;
- updates the Projection and Variance views;
- updates the projection-import date; and
- **does not overwrite any Budget values**.

> **Review before refresh:** refresh is available only while the budget is Draft. Record why the source projection changed and review all resulting variances after refreshing.

9.8 Budget workflow

Status	What it means	Available action
Draft	The budget can be edited and its projection comparison can be refreshed.	Save and Submit.
Submitted	The preparer has completed the draft. Values are read-only while approval is pending.	A Tenant Owner or Tenant Administrator may Approve.
Approved	The agreed budget is locked as the assessment's approved management plan.	A Tenant Owner or Tenant Administrator may Reopen it as Draft when a controlled revision is required.

Every status change is recorded in budget status history with the user and date. Concurrent changes are protected: if another user changes the budget after it was opened, reload the page before saving again.

9.9 Permissions

Tenant role	Budget access
Tenant Owner / Tenant Administrator	Create, edit and submit authorized assessment budgets; approve and reopen submitted or approved budgets.
Branch Manager	Create, edit and submit budgets only for assessments within the manager's authorized scope.
Viewer	Read budgets for authorized assessments but cannot create, edit, submit, approve or reopen them.

Access is checked by the service for every operation. Hiding a button in the page is not the only security control.

9.10 Recommended budget review

1. Confirm the budget name, assessment case and Month 1 start date.
2. Compare Budget with Projection and explain material variances.
3. Review gross profit, EBITDA and profit before tax by month.
4. Check that cash receipts and payments are consistent with the profit assumptions.

5. Review the lowest monthly cash position, not only Month 12 closing cash.
6. Resolve or document funding shortfalls and loss-making months.
7. Save the draft, perform a final Projection and Variance review, and then submit.
8. Approve only after assumptions, evidence, risks and management actions have been reviewed.

9.11 Important limitations

- The first release supports one active 12-month budget per assessment.
- The budget is a planning record, not an accounting ledger or replacement for actual management accounts.
- Budget versus actual reporting is not populated automatically in this release.
- Editing revenue does not automatically rebuild receipts, VAT, stock purchases or working-capital timing inside the budget.
- Refreshing the projection preserves budget values by design; users must review the resulting variances.
- Approving a budget does not change or approve the underlying assessment.
- Displayed amounts use two decimal places while the database retains four-decimal calculation precision.

Recommended overall review sequence

1. Verify the opening date and balances.
2. Check income classification and monthly Sales.
3. Confirm debtor, creditor and stock behaviour.
4. Review expenses, VAT and wages.
5. Reconcile loans and assets to supporting statements.
6. Review warnings, document sources and confirm the baseline.
7. Run sensitivity scenarios and record the principal risks and mitigations.
8. Create the assessment budget, document material variances and review cash requirements.
9. Submit and approve the budget through the authorized workflow.